

Xinyu Li

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RESEARCH INTERESTS	High Energy Astrophysics, Magnetars, Magnetohydrodynamics, Force-Free Electrodynamics, Plasma Astrophysics, Cosmology, Large Scale Structure, Dark Matter, Galaxy Formation	
EDUCATION	Columbia University , New York, NY, USA Ph.D., Physics, August 2019 • Thesis Topic: <i>Magnetodynamics Inside and Outside Magnetars</i> • Advisor: Prof Andrei Beloborodov University of Waterloo , Waterloo, ON, Canada M.Sc, Theoretical Physics, June 2013 • Thesis Topic: <i>Towards an Investigation of Turbulence in High Dimensions</i> • Advisor: Prof Luis Lehner & Prof Robert Myers The University of Hong Kong , HKSAR, China B.Sc (First Class Honor), Physics and Mathematics (Double Major), June 2012 Tsinghua University , Beijing, China Physics and Mathematics, August 2008 - June 2009	
EXPERIENCES	• Assistant Professor Tsinghua University • Postdoctoral Fellow Canadian Institute for Theoretical Astrophysics • Postdoctoral Fellow Perimeter Institute	2023/11- 2019/9-2023/8 2019/9-2022/8
AWARDS	• Jeffrey L. Bishop Fellowship, CITA • Joseph C. Pfister Fellowship, Columbia University • Dean's Fellowship, GSAS, Columbia University • Perimeter Scholars International Scholarship, Perimeter Institute	2021 2018 2013 2012
INVITED TALKS	• Colloquium, T.D.Lee Institute • YTP, Multi-Messenger Astrophysics in the Dynamic Universe • PKU PhD student excellence forum • 3rd FRB and Astrophysics Conference • 2nd Shen Kuo Conference of Chinese Astronomical Society • 1st Tsinghua Workshop on Gravitational Physics • 32nd Texas Symposium on Relativistic Astrophysics • GSI, EMMI+IReNA Workshop "Remnants of neutron-star mergers" • 23rd International Conference on General Relativity and Gravitation • INT, University of Washington, The r-process workshop • Colloquium, Department of Astronomy, Tsinghua University • Colloquium, KIAA, Peking University • Max Planck Institute for Gravitational Physics • North American Einstein Toolkit Workshop • MIAPP High-Energy Plasma Phenomena in Astrophysics	June 2026 Feb 2026 Sept 2025 June 2025 Nov 2024 June 2024 Nov 2023 Oct 2022 July 2022 May 2022 Jan 2022 Dec 2021 Sep 2021 July 2021 July 2021

	• Department of Astronomy, Columbia University	March 2021
	• Perimeter Institute	March 2021
	• KIAS Workshop on Cosmology and Structure Formation	Nov 2020
	• KIAA, Peking University	Oct 2020
	• Department of Astronomy, Tsinghua University	Sep 2020
	• T.D.Lee Institute	May 2020
	• Department of Astronomy, Nanjing University	Jan 2020
	• Department of Astronomy, HUST	Jan 2020
	• Department of Astronomy, Shanghai Jiaotong University	Sep 2020
	• Perimeter Institute	Nov 2018
	• TAC, University of California, Berkeley	Oct 2018
	• KIPAC, Stanford University	Oct 2018
	• TAPIR, California Institute of Technology	Oct 2018
	• CCA, Flatiron Institute	Oct 2018
	• Department of Astrophysical Sciences, Princeton University	Sep 2018
	• Department of Physics, the University of Hong Kong	Aug 2018
SERVICES	• Tsinghua DoA Teaching Committee • Tsinghua DoA Colloquium Committee • CITA Visitor Committee • CITA Cosmology Discussion	
NUMERICAL CODES	Developer: ENZO , FFE-WENO , Einstein Toolkit Experienced user: TRISTAN-MP , RAMSES	
JOURNAL REVIEWER	Physical Review Letters Physical Review D The Astrophysical Journal Letters The Astrophysical Journal Journal of High Energy Physics Journal of Cosmology and Astroparticle Physics Science Bulletin SCIENCE CHINA Physics, Mechanics & Astronomy Physics of Fluids Astrophysics and Space Science Physical Letter B New Astronomy Geophysical and Astrophysical Fluid Dynamics	

PUBLICATIONS

1. Tianshu Wu, **Xinyu Li** and Yangyang Cai, *Rapid Energy Dissipation by Colliding Waves in Strongly Magnetized Plasmas*, arXiv:2608.29365 (2026)
2. Jixuan Yang, Lile Wang, **Xinyu Li** and Rixin Li, *From Inspiral to Expansion: The Wake-Driven Torque on Binary Black Holes in Gaseous Medium*, arXiv:2608.27970 (2026)
3. Yiheng Liu, **Xinyu Li**, *Tidal Heating of Stellar Clusters in Fuzzy Dark Matter Halos*, arXiv:2604.26393 (2026)
4. Jixuan Yang, Lile Wang, **Xinyu Li**, Meng Sun, Long Wang and Rainer Spurzem, *The effects of star-gas interactions on binary evolution in open clusters*, arXiv:2601.01597 (2026)
5. Hong-Xuan Jiang, **Xinyu Li**, Yosuke Mizuno, Ziri Younsi, Christian M. Fromm, *Electromagnetic Signatures of Supermassive Binary Black Holes. I. Thermal Synchrotron, Self-Lensing Flares, and Jet Precession*, The Astrophysical Journal, 1006, 2, 128 (2026)
6. **Xinyu Li**, Houyi, Sun, Yuan-Chuan Zou, and Huan Yang, *Micro-Tidal Disruption Events at Galactic Centers*, The Astrophysical Journal, 996, 2, 120 (2026)
7. Sean M. Ressler, Luciano Combi, Bart Ripperda, **Xinyu Li**, *Black Hole Collisions With Thin Accretion Disks: OJ 287 and Small-Mass-Ratio Supermassive Black Hole Binary Candidates*, The Astrophysical Journal Letters, 993, 1, L22 (2025)
8. Ge, M., Ji, L., Taverna, R., et al., *Physics of strong magnetism with eXTP*, Science China Physics, Mechanics, and Astronomy, 68, 11, 119505 (2025)
9. Jens F. Mahlmann, Miguel . Aloy, **Xinyu Li**, *Force-free wave interaction in magnetar magnetospheres: Computational modeling in axisymmetry*, The Astrophysical Journal, 972, 2, 139 2024
10. Sean M. Ressler, Luciano Combi, **Xinyu Li**, Bart Ripperda, and Huan Yang, *Black Hole-Disk Interactions in Magnetically Arrested Active Galactic Nuclei: General Relativistic Magnetohydrodynamic Simulations Using A Time-Dependent, Binary Metric*, The Astrophysical Journal 967, 1, 70 (2024)
11. Axel Widmark, Tomer D. Yavetz and **Xinyu Li**, *Fuzzy dark matter dynamics in tidally perturbed dwarf spheroidal galaxy satellites*, Journal of Cosmology and Astroparticle Physics, 03, 052 (2024)
12. Jie-Shuang Wang, **Xinyu Li**, Zigao Dai and Xuefeng Wu, *An Intermediate-field Fast Radio Burst Model and the Quasi-periodic Oscillation*, Research in Astronomy and Astrophysics, 23, 3 (2023)
13. Alexander Y. Chen, Yajie Yuan, **Xinyu Li** and Jens F. Mahlmann, *Propagation of a Strong Fast Magnetosonic Wave in the Magnetosphere of a Neutron Star*, arXiv:2210.13506 (2022)
14. Mihir Kulkarni, Eli Visbal, Greg L. Bryan, and **Xinyu Li**, *If dark matter is fuzzy, the first stars form in massive pancakes*, The Astrophysical Journal Letters, 941, 1 (2022)
15. Alexander Laroche, Daniel Gilman, **Xinyu Li**, Jo Bovy and Xiaolong Du, *Quantum fluctuations masquerade as halos: Bounds on ultra-light dark matter from quadruply-imaged quasars*, Monthly Notices of the Royal Astronomical Society, 517, 2 (2022)
16. Lile Wang, **Xinyu Li**, *Dynamical Effects of Colliding Outflows in Binary Systems*, The Astrophysical Journal, 932, 108 (2022)

17. Alexander Y. Chen, Yajie Yuan, Andrei M. Beloborodov and **Xinyu Li**, *Relativistic Alfvén Waves Entering Charge Starvation in the Magnetospheres of Neutron Stars*, The Astrophysical Journal, 929, 1, 31
18. Tomer D. Yavetz, **Xinyu Li**, Lam Hui, *Construction of Wave Dark Matter Halos: Numerical Algorithm and Analytical Constraints*, Physical Review D, 105, 023512 (2022)
19. **Xinyu Li**, Andrei M. Beloborodov, Lorenzo Sironi, *Fast dissipation of Colliding Alfvén Waves in a Magnetically Dominated Plasma*, The Astrophysical Journal, 915, 101 (2021)
20. **Xinyu Li** and Daniel Siegel, *Neutrino Fast Flavor Conversions in Neutron-star Post-Merger Accretion Disks*, Physical Review Letters, 126, 25, 251101 (2021)
21. Neal Dalal, Jo Bovy, Lam Hui, **Xinyu Li** *Don't cross the streams: caustics from Fuzzy Dark Matter*, Journal of Cosmology and Astroparticle Physics, 03, 076 (2021)
22. **Xinyu Li**, Lam Hui, Tomer D. Yavetz, *Oscillations and Random Walk of the Soliton Core in a Fuzzy Dark Matter Halo*, Physical Review D, 103, 023508 (2021)
23. Lam Hui, Austin Joyce, Michael J. Landry, **Xinyu Li** (alphabetical order), *Vortices and waves in light dark matter*, Journal of Cosmology and Astroparticle Physics, 01, 011 (2021).
24. Brummel-Smith, C., Bryan, G., Butsky, I., et al. *ENZO: An Adaptive Mesh Refinement Code for Astrophysics*, Journal of Open Source Software, 4, 1636.
25. **Xinyu Li**, Philip Chang, Yuri Levin, Christopher D. Matzner, Philip J. Armitage, *Simulation of a Compact Object with Outflows Moving Through a Gaseous Background*, Monthly Notices of the Royal Astronomical Society, 494, 2 (2020)
26. F.Y. Wang, **Xinyu Li**, D. O. Chernyshov, C.Y. Hui, G.Q. Zhang and K.S. Cheng, *Consequences of Energetic Magnetar-like Outbursts of Nearby Neutron Stars: ^{14}C Events and the Cosmic Electron Spectrum*, The Astrophysical Journal, 887, 2, (2019)
27. Lam Hui, Daniel Kabat, **Xinyu Li**, Luca Santoni and Sam S. C. Wong (alphabetical order), *Black hole hair from scalar dark matter*, Journal of Cosmology and Astroparticle Physics, 06, 038 (2019)
28. **Xinyu Li**, Jonathan Zrake and Andrei Beloborodov, *Dissipation of Alfvén Waves in Relativistic Magnetospheres of Magnetars*, The Astrophysical Journal, 881, 1, (2019)
29. **Xinyu Li**, Lam Hui and Greg Bryan, *Numerical and Perturbative Computations of the Fuzzy Dark Matter Model*, Physical Review D, 99, 63509, (2019)
30. **Xinyu Li**, Yuri Levin and Andrei Beloborodov, *Magnetar Outbursts from Avalanches of Hall Waves and Crustal Failures*, The Astrophysical Journal, 189, 12, (2016)
31. Andrei Beloborodov and **Xinyu Li**, *Magnetar Heating*, The Astrophysical Journal, 261, 20, (2016)
32. **Xinyu Li** and Andrei Beloborodov, *Plastic Damping of Alfvén Waves in Magnetar Flares and Delayed Afterglow Emission*, The Astrophysical Journal, 815, 25, (2015)

33. **Xinyu Li** , Fayin Wang and K.S. Cheng, *Gravitational Effects of Condensate Dark Matter on Compact Stellar Objects*, Journal of Cosmology and Astroparticle Physics, 10, 031, (2012)
34. **Xinyu Li** , T. Harko and K.S.Cheng, *Condensate Dark Matter Stars*, Journal of Cosmology and Astroparticle Physics, 06, 001, (2012)